

TW10: Importing Data in Python

Five challenges | 20 marks each | 100 marks total

Student name	
Student ID	
Class / group	
Submission date	

Weekly purpose

Import text, CSV, Excel, JSON, trusted pickle, and compressed files while documenting structure, options, and provenance.

SUBMISSION RULES

- Complete work in the corresponding Jupyter notebook.
- Write your own code and explanations; no solutions are supplied.
- Keep supplied data unchanged and work on copies.
- Restart the kernel and run all cells before submission.
- Submit the notebook with outputs visible and assistance acknowledged.

MARKING RUBRIC FOR EACH CHALLENGE

Assessment area	Marks
Correct method and working code	8
Validation and quality checks	4
Explanation and interpretation	4
Reproducibility and presentation	4
Total per challenge	20

Challenge 1: Inventory files and decode text

Scenario

You received a data directory with two text encodings and several tabular formats.

INSTRUCTIONS

1. List every supplied file with name, suffix, and byte size.
2. Read the UTF-8 notes and Latin-1 notes with explicit encodings.
3. Count lines and verify that accented characters and the pound sign are preserved.
4. Explain why `errors='ignore'` is not an acceptable default.

REQUIRED SUBMISSION

- A sorted file inventory DataFrame.
- Correct text output from both files.
- Two encoding-validation checks and a short risk note.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 2: Create a CSV import contract

Scenario

The sales CSV will be used in a report and must satisfy an explicit schema.

INSTRUCTIONS

1. Import only `order_id`, `order_date`, `region`, `units`, and `unit_price`.
2. Set `order_id` to string, `region` to category, and parse the date.
3. Validate required columns, eight rows, unique order IDs, non-negative units, and positive prices.
4. Calculate revenue and summarise it by region.

REQUIRED SUBMISSION

- A typed five-column DataFrame.
- At least five readable assertions.
- A sorted regional revenue table.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 3: Import and join Excel worksheets

Scenario

An Excel workbook contains Sales and Regions sheets that must be checked and combined.

INSTRUCTIONS

1. Inspect sheet names before reading data.
2. Read both sheets by name and validate their expected columns.
3. Left-merge manager and target onto sales using region.
4. Use merge validation and an indicator to find unmatched rows.

REQUIRED SUBMISSION

- A worksheet inventory with shapes.
- A combined table retaining all sales rows.
- A match-quality report and explanation of many-to-one validation.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 4: Flatten JSON and document pickle safety

Scenario

Nested customer JSON must become a rectangular table. A locally supplied pickle is also available.

INSTRUCTIONS

1. Load JSON with the standard library and inspect its top-level structure.
2. Flatten contact fields with `pandas.json_normalize`.
3. Select `customer_id`, `name`, `active`, `city`, and `postcode`.
4. Load only the trusted supplied pickle and compare its shape with `sample_sales.csv`.
5. Write a clear warning explaining why unknown pickle files must not be opened.

REQUIRED SUBMISSION

- A five-column flattened customer table.
- JSON structure and row-count checks.
- A pickle-versus-CSV comparison and security warning.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 5: Process compressed data and write provenance

Scenario

The same sales data appears as plain CSV, gzip, and a CSV member inside a ZIP archive.

INSTRUCTIONS

1. Read the gzip CSV directly with pandas.
2. Inspect ZIP member names and open only sales_inside.csv.
3. Read the plain CSV in chunks of two rows and accumulate row count, units, and revenue.
4. Verify equivalent values across representations.
5. Create a provenance log recording path, format, member or encoding, options, rows, and columns.

REQUIRED SUBMISSION

- Three validated imports plus one chunked aggregation.
- Equality or discrepancy checks.
- A complete provenance table.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Final submission checklist

- ☐ All five challenges are complete.
- ☐ The notebook runs from top to bottom without errors.
- ☐ Every output is clearly labelled.
- ☐ Important results include written explanations.
- ☐ Validation checks are included.
- ☐ Supplied raw data has not been overwritten.
- ☐ Sources and assistance are acknowledged.

Student declaration

I confirm that this submission is my work, that I have acknowledged sources and assistance, and that the displayed outputs were produced by the submitted code.

Signature: _____ Date: _____